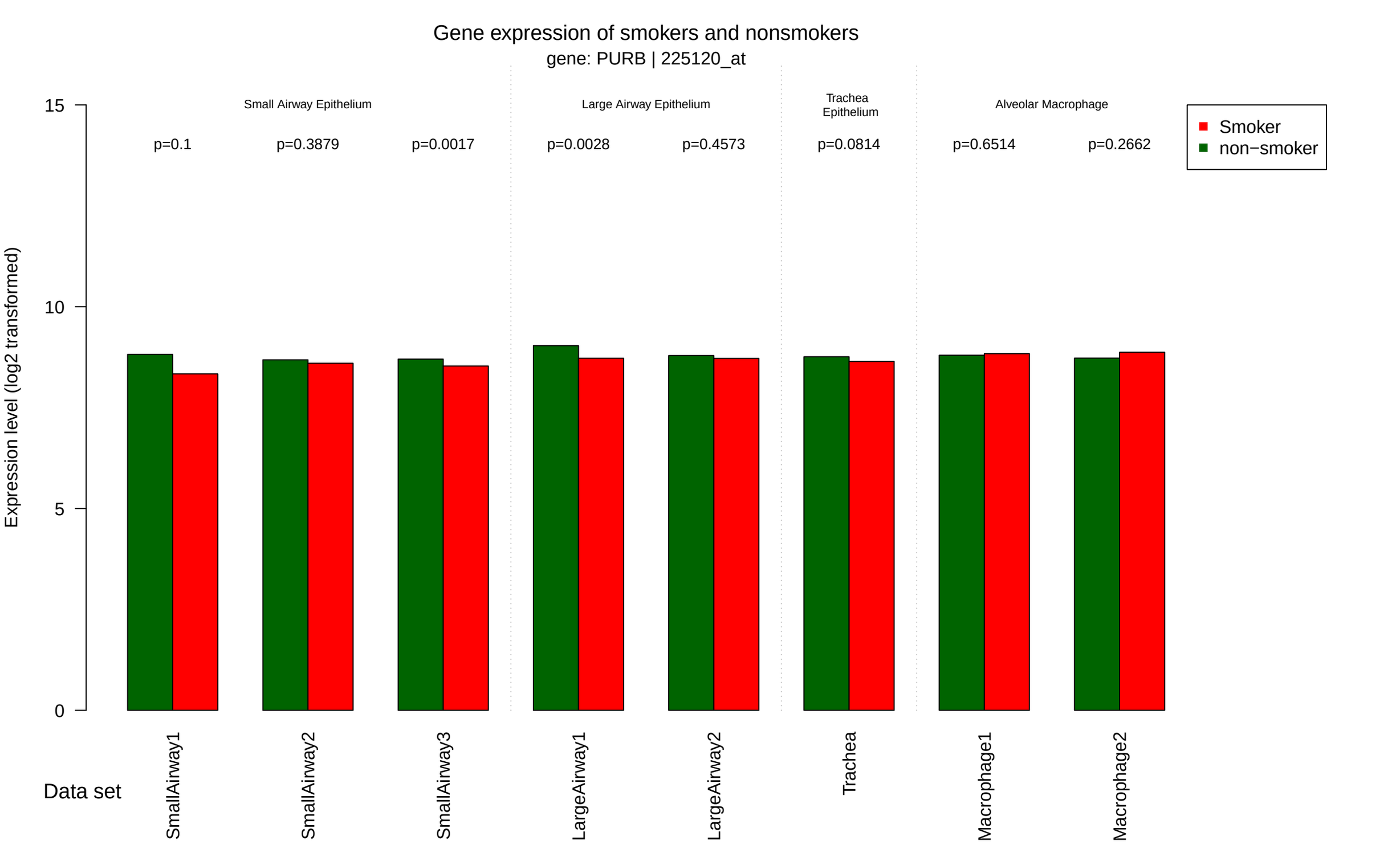
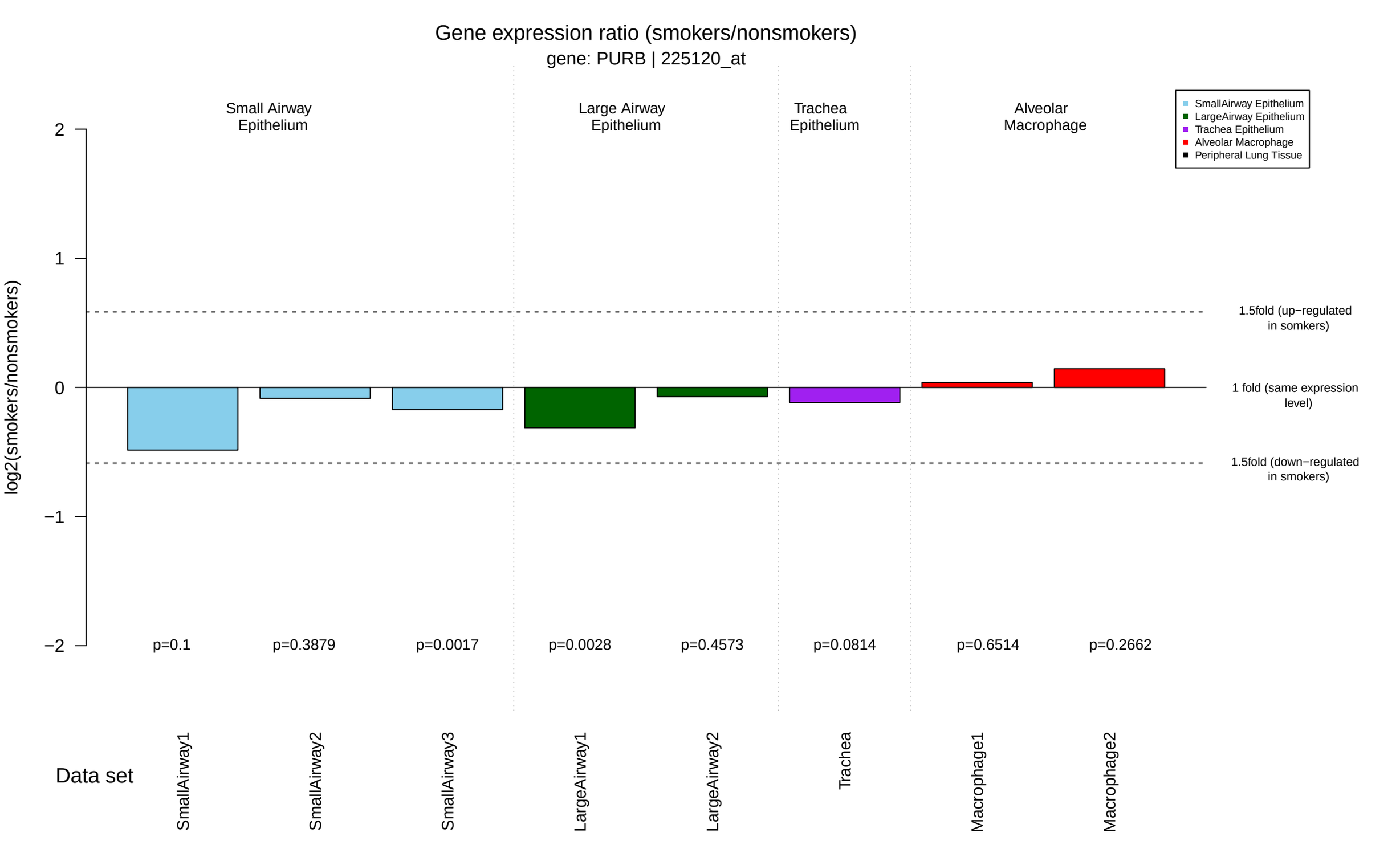
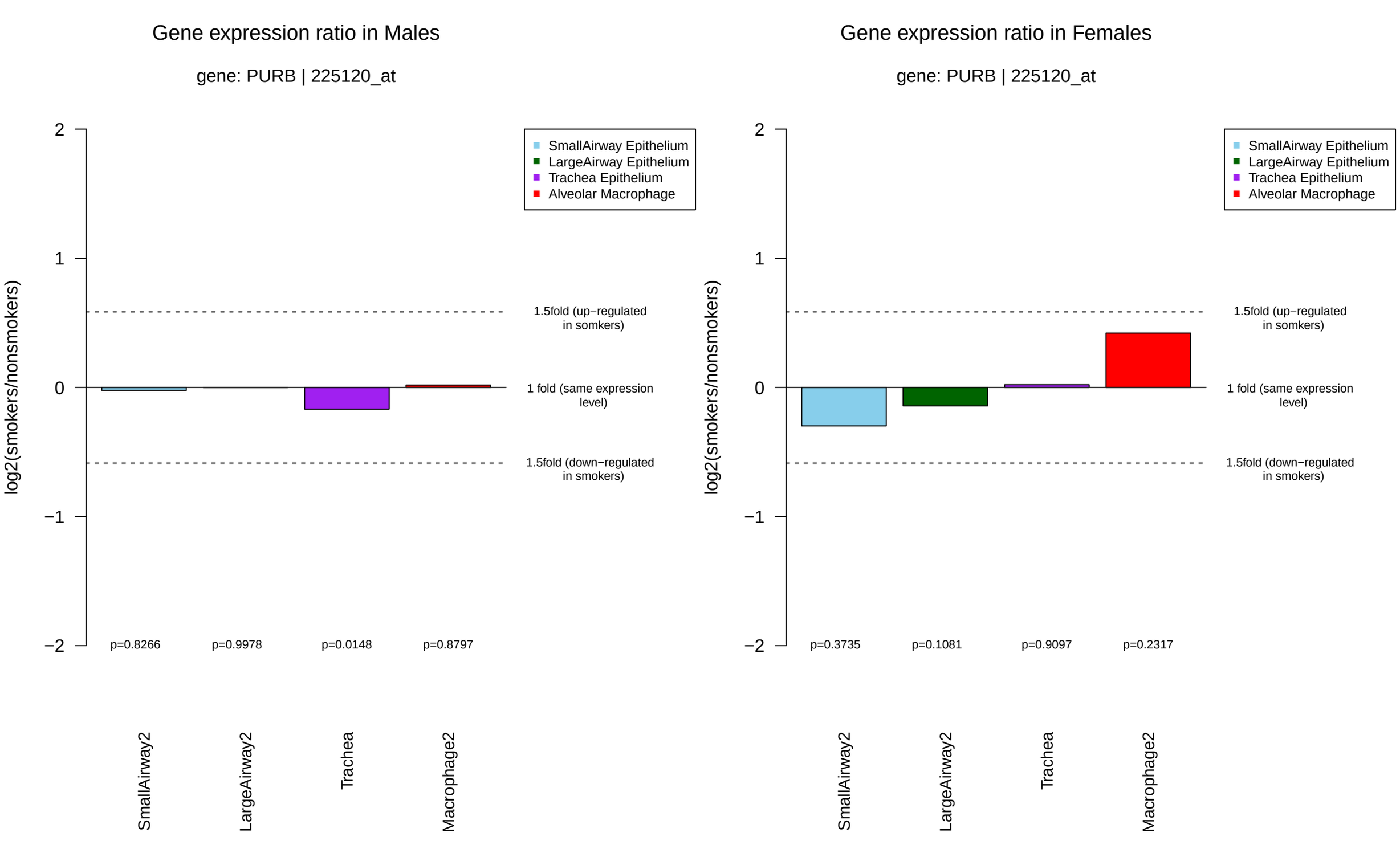
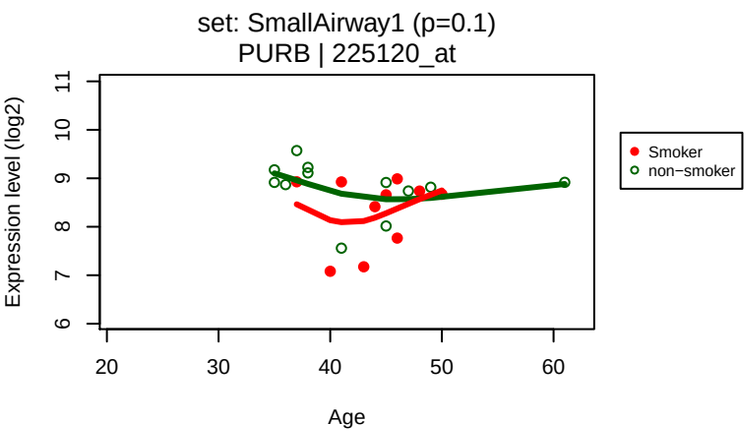
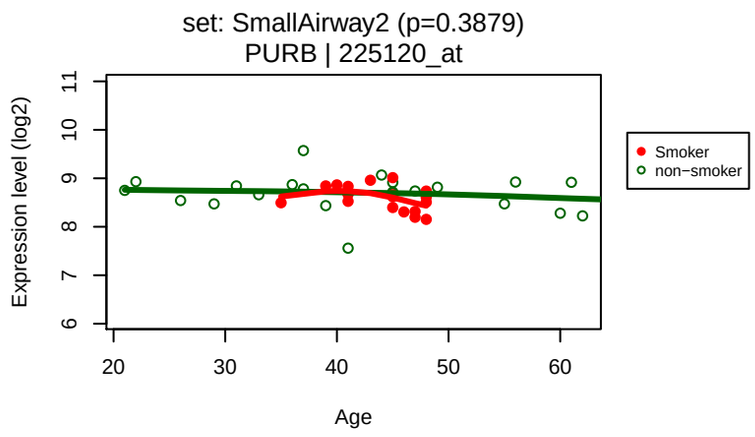
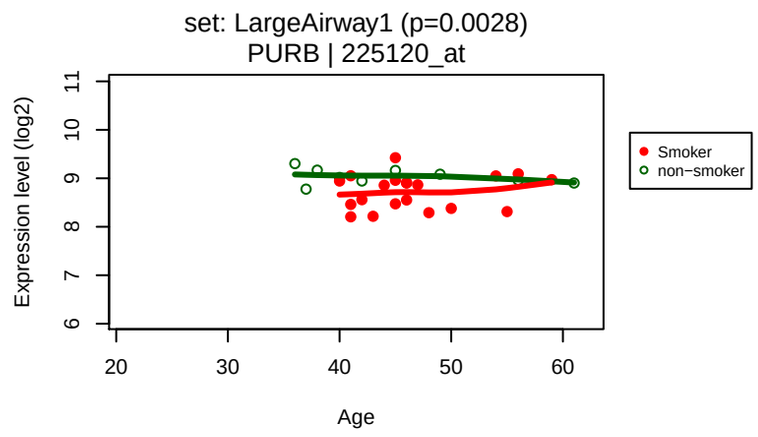

Gene expression of smokers and nonsmokers
gene: PURB | 225120_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1
p=0.3879
p=0.0017
p=0.0028
p=0.4573
p=0.0814
p=0.6514
p=0.2662
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

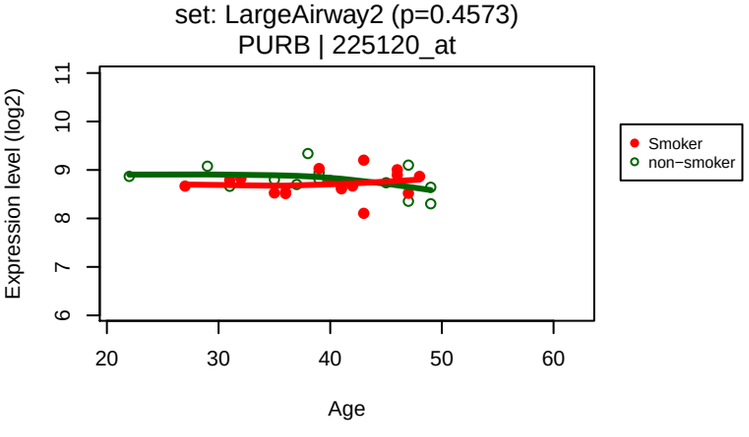

Gene expression ratio (smokers/nonsmokers)
gene: PURB | 225120_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1
p=0.3879
p=0.0017
p=0.0028
p=0.4573
p=0.0814
p=0.6514
p=0.2662
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

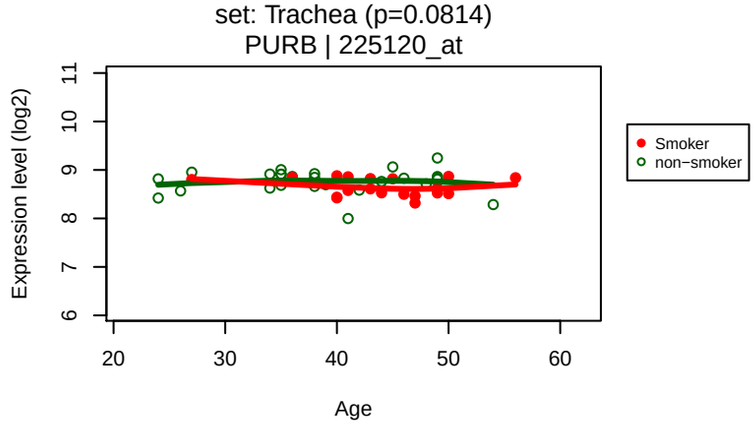

Gene expression ratio in Males
gene: PURB | 225120_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8266
p=0.9978
p=0.0148
p=0.8797
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PURB | 225120_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3735
p=0.1081
p=0.9097
p=0.2317
SmallAirway2
LargeAirway2
Trachea
Macrophage2

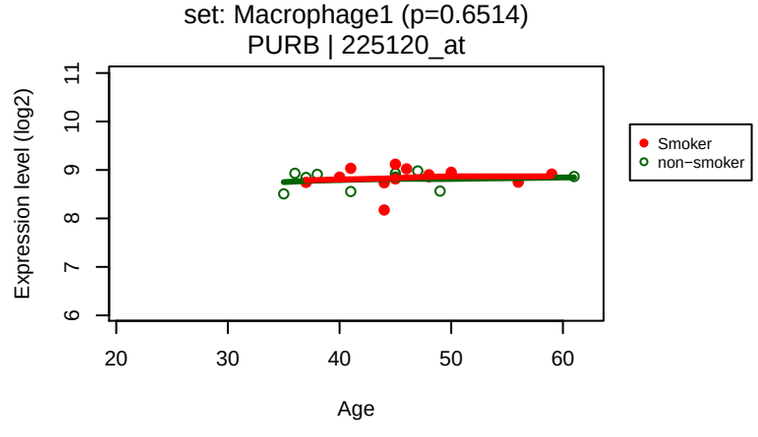

set: SmallAirway1 (p=0.1)
PURB | 225120_at
Expression level (log2)
Age
Smoker
non−smoker

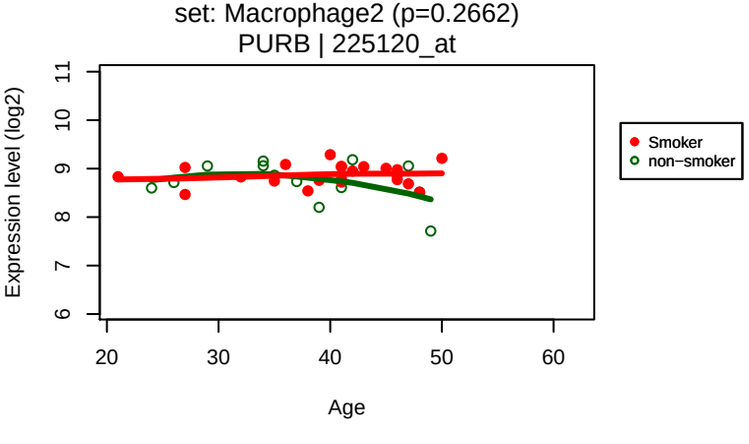

set: SmallAirway2 (p=0.3879)
PURB | 225120_at
Expression level (log2)
Age
Smoker
non−smoker

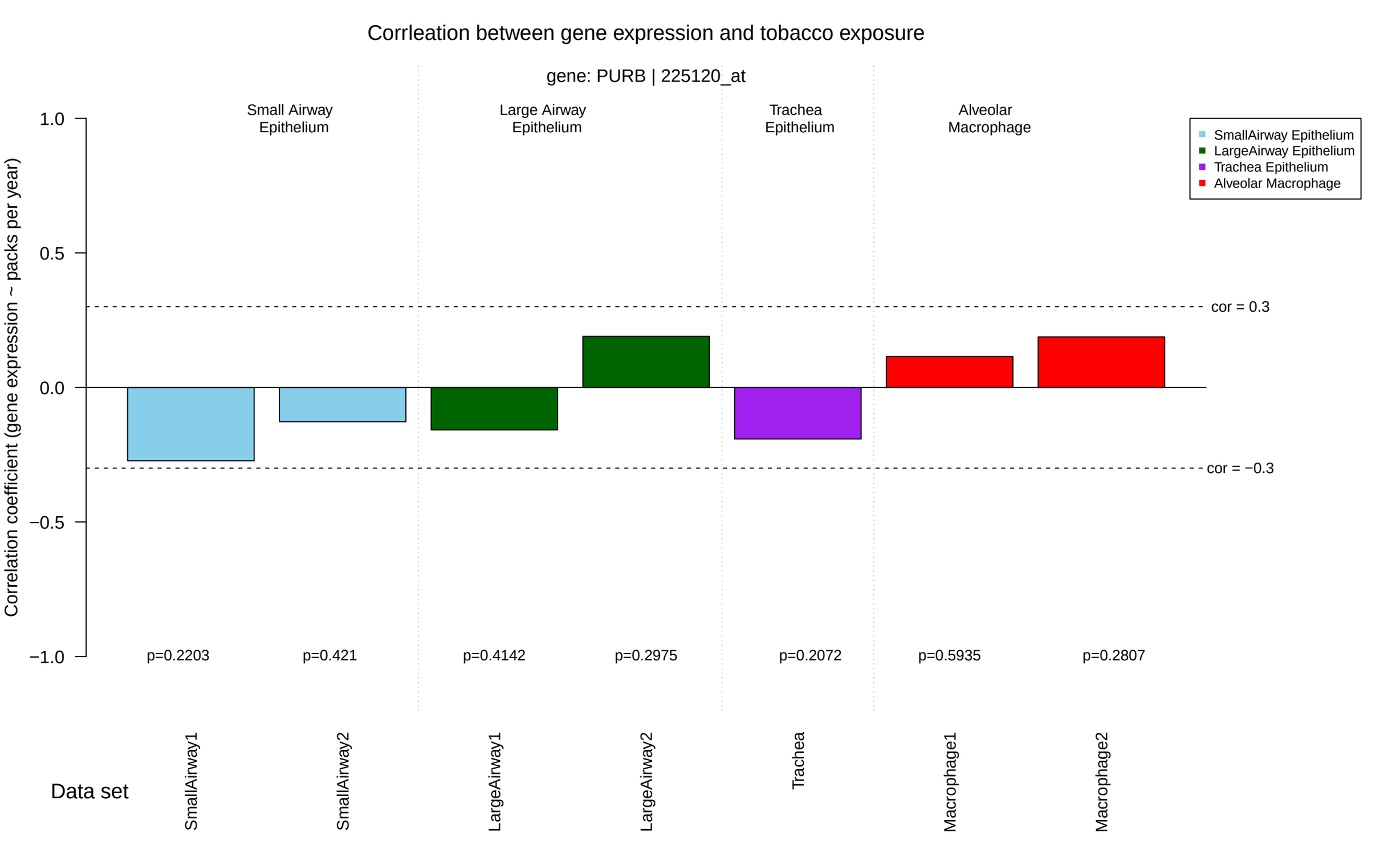

set: LargeAirway1 (p=0.0028)
PURB | 225120_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4573)
PURB | 225120_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0814)
PURB | 225120_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6514)
PURB | 225120_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2662)
PURB | 225120_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PURB | 225120_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2203
p=0.421
p=0.4142
p=0.2975
p=0.2072
p=0.5935
p=0.2807
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2